

## WEEK 1 (DAY 5)

**Modules in Python:** A module is a file containing Python code (functions, variables, and classes) that can be reused in other Python programs using the `import` statement.

Types of Modules

1. Built-in Modules
2. User-defined Modules

### Module Operations

1. Importing an Entire Module: Imports all functions and variables from a module.

```
import math

print(math.sqrt(25))
print(math.factorial(5))

5.0
120
```

2. Importing Specific Functions: Imports only the required functions from a module.

```
from math import sqrt, factorial

print(sqrt(36))
print(factorial(4))

6.0
24
```

3. Importing a Module with an Alias: Assigns a shorter name to the module.

```
import math as m

print(m.pi)
print(m.pow(2,3))

3.141592653589793
8.0
```

4. Using the Random Module:Generates random values.

```
import random

print(random.randint(1,10))
print(random.choice(["Apple", "Banana", "Mango"]))
```

1  
Apple

5. Using the Datetime Module:Works with date and time.

```
import datetime

today = datetime.datetime.now()
print(today)
```

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6. Viewing Available Functions in a Module:Displays all functions and variables of a module.

```
import math

print(dir(math))
```

['\_\_doc\_\_', '\_\_loader\_\_', '\_\_name\_\_', '\_\_package\_\_', '\_\_spec\_\_', 'acos', 'acosh', 'asin', 'asinh', 'atan', 'atan2', 'atanh',

## Local and Global variables

A variable is a named location used to store data. Variables can be local or global depending on where they are declared.

Local Variable:A local variable is declared inside a function and can only be accessed within that function.

```
def display():
    message = "Hello"
    print(message)

display()

Hello
```

Global variable : A global variable is declared outside all functions and can be accessed throughout the program.

```
name = "Khushi"
```

```
def show():  
    print(name)
```

```
show()  
print(name)
```

```
Khushi  
Khushi
```